



# Choupo

The Open Source Chemical Process Simulator

## Explorer Guide

Choupo-dev

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# 1 What the Explorer is

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The Explorer is Choupo's *interactive thermodynamics bench*: pick components, pick a view, and SEE what the selected property models do — instantly, in the browser, with no case authoring at all. It answers the question that comes *before* any simulation: “**what does this thermodynamics look like?**”

It sits deliberately between two other surfaces:

- The **props bench** (`choupoProps` cases) asks the same questions *reproducibly*: a case file, golden-locked numbers, validation blocks. The Explorer is its sketchpad — what you find here, you pin down there.
- The **simulators** consume the models the Explorer displays. A surprise in a flowsheet (an azeotrope, a scaling mineral, a runaway  $\gamma$ ) is usually visible in the Explorer in thirty seconds.

Everything the Explorer draws is computed by the SAME C++ engine (compiled to WebAssembly) with the SAME curated catalogue (`data/standards/`) the solvers use. There is no “display model”: if the Explorer shows an azeotrope at  $x = 0.89$ , the distillation column will hit it at  $x = 0.89$ .

## 2 The workflow

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1. **Pick components** in the left panel. The list is the live catalogue; each card shows the data the component actually carries (criticals,  $P^{\text{sat}}$ ,  $c_p$ , formation — and what it lacks). Case-local components (from an open case's **constant**/) appear alongside, marked.
2. **Pick a view** (the tabs enable themselves when their component-count and data requirements are met; a disabled tab tells you WHY it is disabled).
3. **Pick the method** where more than one applies — activity model, equation of state,  $P^{\text{sat}}$  correlation. Overlaying two methods on the same axes is the fastest modelling lesson Choupo offers: the gap between the curves IS the modelling uncertainty.
4. **Export**: every view is a Plotly figure (zoom, hover, save PNG) backed by numbers you can carry into a props case for the golden-locked version.

## 3 The views, one by one

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### 3.1 Property vs $T/P$ (scan)

Any pure-component property —  $P^{\text{sat}}$ , liquid density,  $c_p$ , viscosity, enthalpy — swept over temperature or pressure, for one or many components. Use it to: check a correlation's range before trusting it in a case; compare components; spot the nonsense that extrapolation produces outside a fit's window (the guide's recurring lesson: a correlation is DATA, and data has a domain).

### 3.2 Pure phase diagram ( $P$ – $T$ )

The full phase map of one substance: sublimation, melting and vapour-pressure lines meeting at the triple point, ending at the critical point. Expect water's negatively-sloped melting line (the anomaly), and the supercritical region where the liquid/vapour distinction dissolves.

### 3.3 Binary boiling envelope ( $T$ – $x$ – $y$ )

The bubble and dew curves of a binary at fixed  $P$ . This is where azeotropes announce themselves: the curves touch, and no distillation can cross the touching point. Switch the activity model (ideal  $\rightarrow$  NRTL  $\rightarrow$  UNIFAC) and watch the envelope bend — ideal solutions cannot azeotrope; real ones do.

### 3.4 $\gamma(x)$ — activity coefficients

The raw non-ideality of a binary:  $\gamma_1, \gamma_2$  against composition, with the infinite-dilution values  $\gamma^\infty$  at the edges.  $\gamma > 1$ : the pair dislikes mixing (positive deviation, minimum-boiling azeotropes);  $\gamma < 1$ : association (maximum-boiling). The Gibbs–Duhem constraint ties the two curves together — one cannot bend without the other answering.

### 3.5 McCabe–Thiele (distillation)

The classic  $x$ – $y$  construction: equilibrium curve, operating lines,  $q$ -line, stages stepped off graphically. Change reflux and watch the stage count respond; push the specs toward an azeotrope and watch the pinch strangle the staircase. The fastest intuition-builder for distillation that exists — and it runs on the same  $K$ -values the rigorous column uses.

### 3.6 Binary flash ( $x$ – $y$ + lever rule)

One flash, drawn: feed, vapour and liquid on the equilibrium diagram with the lever rule made visible. Use it to understand what a single equilibrium stage can and cannot do.

### 3.7 Binary and ternary LLE

Liquid–liquid splitting from the mixing Gibbs energy:  $g_{\text{mix}}(x)$  with the common-tangent construction (binary), and the ternary solubility envelope with tie-lines (UNIFAC-predicted). Expect the tangent points — not the minima — to define the coexisting phases; that distinction is the whole thermodynamics of demixing.

### 3.8 Ternary boiling surface

$T_{\text{bubble}}$  over the composition triangle: valleys are distillation boundaries, and azeotropes sit at the stationary points.

### 3.9 Psychrometric chart

Humid air: humidity vs temperature with saturation, wet-bulb and enthalpy structure — the working chart for dryers and HVAC, computed from the same gas-phase thermodynamics as everything else.

### 3.10 Scaling (SI vs recovery)

The membrane/brine screening view: concentrate a water composition over recovery and watch each mineral's saturation index climb toward  $SI = 0$  (the precipitation line). The Davies-vs-Pitzer method switch is the lesson: at high ionic strength the two activity models give DIFFERENT scaling verdicts, and choosing one is an engineering decision.

### 3.11 Equilibrium map (Gibbs)

The chemical-reaction view: pick two or more gas-phase species (with plain formulas — the atom matrix is derived from them, glass-box) and a feed, and get iso-lines of a product's equilibrium mole fraction over the  $T \times P$  plane. The whole point is the tension it makes visible: for ammonia ( $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ ) the equilibrium optimum sits cold and high-pressure, yet the labelled industrial window sits hot — where the catalyst can work. Answers the beginner's real confusion first (“where do I write the reaction?” — you don't; Gibbs redistributes the ATOMS you fed, shown as a live element balance). *Click any cell* for its full equilibrium composition and the ready-to-run `gibbsReactor` case dict — the map is a case generator. The *advanced* toggle exposes the temperature-approach  $\Delta T$  (the empirical closeness-to-equilibrium knob commercial tools hide); it snaps rather than animates, and ghosts

the  $\Delta T = 0$  contours underneath so you read the before/after, not a motion. Unconverged cells are marked, never interpolated.

### 3.12 Steam tables (IF97)

Saturation and superheat properties of water from IAPWS-IF97 —  $h_f, h_g, h_{fg}$ , entropies, volumes — the backbone of every steam and power-cycle case, browsable.

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## 4 Provenance and honesty rules

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The Explorer inherits the project's credo:

- **Every number has a source.** Component cards show where each value came from; estimates carry their UNVALIDATED badge; nothing is silently filled in.
- **Disabled means explained.** A view that cannot run tells you which requirement failed (component count, missing VLE data, missing UNIFAC groups) — never a dead button.
- **The Explorer never writes.** It reads the catalogue and the open case; promoting data, fitting parameters and locking goldens belong to the props bench and the curation tools. Sketch here, commit there.

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## 5 From Explorer to case: the hand-off

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A typical session: you notice in the  $T$ - $x$ - $y$  view that your solvent pair azeotropes at 78% — so the simple column in your flowsheet cannot deliver 99% purity. The hand-off is: (1) reproduce the envelope as a props case (`propertyScanBinary`) so the finding is golden-locked; (2) switch the flowsheet to the azeotrope-aware design (pressure-swing or entrainer); (3) cite the Explorer finding in the case header. The Explorer is where you LEARN; cases are where knowledge is kept.